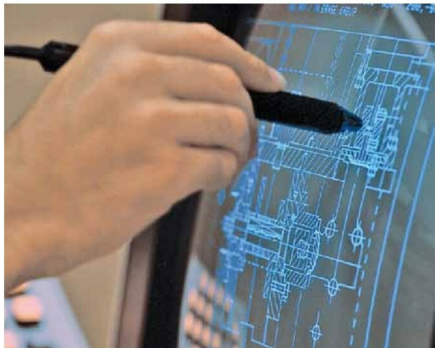


WHAT IT TAKES TO BE A 'GOOD' EMBEDDED ENGINEER?

Embedded systems are becoming more and more complex and so is the skillset of an embedded engineer. Gary Stringham and Robert Oshana, speakers at Embedded System Conference 2010, spoke on how the market demand and the culture are driving the skillset requirements of an embedded engineer



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The definition of embedded system has broadened over the years. The initial definition said that an embedded system is anything that is application-specific, but, today, an embedded device like a mobile phone can do much more than just calling—it can play FM radio, click pictures and so on.

"Whatever be the additional feature, an embedded system will always be resource-constrained. The challenge to design with limited power, limited processing speed, limited memory and

limited input/output (I/O) will always be there," says Robert Oshana, director of software research and development, networking and multimedia group, Freescale Semiconductor.

Embedded engineers always need to optimise what they are doing, unlike desktop engineers who can push more memory or more processing speed. If we look around, there are many examples of pure embedded devices, and of larger systems built using multiple embedded sub-systems.

Here are some suggestions for those who aspire to become an embedded engineer and also for those who

are already an embedded engineer.

Be an all-rounder

The embedded world is outside the desktop world, where pressing the 'on' button will not boot the system. An embedded engineer needs to be an all-rounder. "He has to know much more than just writing the code. He has to understand the hardware, how it interacts with the environment, how to use controls, sensors and actuators, and how to integrate all together," says Oshana.

Today's education system will teach you either hardware in electrical engineering or software in computer science. Even the computer engineering courses do not cover embedded systems completely. "Embedded engineers need a broader set of educational qualification. Cross-functional curriculum for the embedded discipline tailored to embedded career will be useful," suggests Oshana.

Buy a development board

It's said that if a kid can take a thing apart and then put it back together such that it still works, he will become an engineer. Gary Stringham, president, Gary Stringham & Associates, believes the same as he feels that engineering is a lot about self-learning through experimentation. He suggests, "You need to buy a development board and play with it. Program it, and get one LED to blink. Then program it to blink fast and slow. Learn how to connect it to your computer.